

generator

Hardcoded-content audit — Go generator (`_tools/gen/*.go`)

Read-only, multi-pass audit for user-facing PROSE/COPY baked into code rather than sourced from `data/content/i18n` (`_content/**`, `portfolio.json`, `ui-i18n.json`). Constitution §11.4.140/141/236 + OpenDesign data-driven mandate.

Scope: every non-test `.go` file under `_tools/gen/` (`main.go`, `data.go`, `i18n.go`, `seo.go`, `markdown.go`, `home.go`, `portfolio.go`, `product.go`, `shells.go`). `*_test.go` are assertions, not emitted output → out of scope (noted below). `_tools/pdf/*` is shell/JSON, not Go → out of scope of this file.

Date: 2026-08-07. No files were modified.

Summary counts

Severity	Count	Findings
HIGH	1	F1 (creativeHeadings — ~40 English section headings)
MED	4	F2, F3 (untranslated alt text ×2), F4 (hardcoded tier stat “3”), F5 (English meta-desc fallback suffix)
LOW	3	F6 (dead portfolioLede/ portfolioSummary prose consts), F7 (hardcoded “© 2026” year ×3), F8

Severity	Count	Findings
		(JSON-LD operatingSystem/ applicationCategory literals)

Genuine violations to fix: **8**. False-positives explicitly cleared: see “Not violations” — the `to(...)` English fallbacks are all backed by present keys and never fire.

HIGH

F1 — `markdown.go:210–271` `creativeHeadings` map (+ `editorialHeading` 275–284)

Literal (truncated):

"why we built it": {"The problem we set out to solve", "Why this exists", "What drove us to build it", "The itch we had to scratch", "Origin story"}, "why it's a game-changer": {"Why it changes the game", ...}, ... (~40 English section-heading strings across 10 canonical keys).

Why a violation: these are visitor-facing **section headings** rendered onto every product detail page (via `renderProductAccordion` → `editorialHeading`). They are English prose defined entirely in Go with no data/i18n source. On the EN pages they *replace* the author’s markdown `##` heading with a reworded English variant; on localized pages the lookup key (English, lowercased) does not match the translated heading, so `editorialHeading` returns the author’s translated heading verbatim — meaning the reworded copy exists **only** in code and **only** in English, and can never be translated or reviewed through the normal pipeline. This is the single largest body of hardcoded copy in the generator.

Severity: HIGH — substantial visitor-facing prose, code-only, untranslatable.

Remediation: move the heading-variant table out of code into a data/i18n surface. Two viable shapes: (a) an i18n table (e.g. `ui-i18n.json` keys `prod.head.<canonicalKey>.<n>`) resolved via `T(lang, ...)` so localized product pages also get the editorial rewording and it goes through reviewer gating; or (b) a `_content/-side` JSON (e.g. `_content/products/heading-variants.json`) loaded at gen time. Per constitution the i18n route (a) is preferred because these are translatable UI copy. Keep the deterministic per-slug hash selection.

MED

F2 — `home.go:196 alt="Portrait of Miloš Vasić" (inside const mvPortrait)`

Why: alt text is user-facing (screen readers / image-off) and is hardcoded English with no i18n key (confirmed: no `alt.*` key exists in `ui-i18n.json`). Emitted on every `milosvasic.ru` home, all languages, always English. Severity: MED (untranslated user-facing label). Remediation: add key e.g. `alt.portrait` to `ui-i18n.json`; convert `mvPortrait` from a `const` to a `func` taking `lang` and interpolate `esc(T(lang, "alt.portrait"))`.

F3 — `home.go:206 alt="Vasic Digital logo" (inside func vdHeroLogo)`

Why: same class as F2 — hardcoded English alt on the `vasic.digital` hero logo, no i18n key. Severity: MED. Remediation: add key e.g. `alt.brandLogo`; pass `lang` into `vdHeroLogo` (currently takes only prefix) and emit `esc(T(lang, "alt.brandLogo"))`. (Note: the small header brand logo at `home.go:332` uses `alt=""` — correct, decorative; not a finding.)

F4 — `portfolio.go:134 hardcoded <div class="od-stat__value">3</div>`

Literal: the middle portfolio hero stat value 3 (the “Tiers” count). Why: a number-as-content baked in code instead of derived from data. The other two stat values on the same block are data-driven (`p.Count`, `helixCount`); only the tier count is a literal. It happens to equal `len(p.Tiers)` today (`portfolio.json` tiers has 3 entries) but will silently go stale if a tier is added/removed — contradicting the data-driven mandate. Severity: MED (stale-risk, data-driven violation). Remediation: interpolate `len(p.Tiers)` (`p` is already in scope in `portfolioMainInner`) with a `%d`, mirroring `p.Count`/`helixCount`.

F5 — `product.go:154 English fallback suffix in meta description`

Literal: `desc := firstNonEmpty(e.Summary, e.Tagline, e.Name+" — "+site.Brand+" product.")` Why: the third arg synthesizes an English meta description (`product.`) baked in code; it feeds `<meta name=description>`, `og:twitter:description`, and the `SoftwareApplication` JSON-LD description. Currently latent — all 33 `portfolio.json` entries have a summary or tagline, so it never fires today — but it is hardcoded, English-only, non-translatable copy waiting to surface. Severity: MED (latent visitor-facing SEO prose;

MED because SEO-facing, though currently unreached).
Remediation: drop the English literal; fall back to an i18n template, e.g. `fmt.Sprintf(T(lang, "prod.desc.fallback"), e.Name, site.Brand)` with the key in `ui-i18n.json`, or simply fall back to the already-localized page title.

LOW

F6 — portfolio.go:20–21 const portfolioLede / const portfolioSummary

Literals:

"A unified, evidence-based portfolio of the Helix family, ..." and "One fleet: large product applications on top of dozens of small, decoupled, independently-tested modules – governed by a shared engineering Constitution ...". Why: full paragraphs of English marketing prose in code. **Dead:** grep shows only the definitions — no references. The live portfolio lede/summary are correctly sourced via `T(lang, "pf.lede") / T(lang, "pf.summary")` (portfolio.go:146,152; keys present for 15 langs). So these consts are vestigial hardcoded copy that predates the i18n migration. Severity: LOW (not emitted) — but they are exactly the anti-pattern the constitution forbids and should be removed to avoid regressions. Remediation: delete both consts.

F7 — Hardcoded copyright year © 2026 (×3)

Locations: portfolio.go:236 (© 2026 %s – %s), product.go:244 (© 2026 %s – %s.), i18n.go:154 ("footer.text": "© 2026 " + site.Brand + ...). Why: the year is content baked in code; it will silently go stale in 2027. The brand and suffix on the same lines ARE data-driven (site.Brand, `T(lang, "footer.suffix")`), so only the literal year is the issue. Not a translation issue (digits), but a data-driven/freshness one. Severity: LOW. Remediation: derive the year (`time.Now().Year()`) or source it from site config; centralize the footer string so all three call sites share one source.

F8 — seo.go:246 JSON-LD operatingSystem/applicationCategory literals

Literals: "applicationCategory": "DeveloperApplication", "operatingSystem": "Cross-platform" (in `softwareApplicationNodeLang`). Why: borderline. These are schema.org structured-data enumerated values, which the audit rules classify as legit code strings (JSON-LD @types/values). They are not rendered as visible prose. operatingSystem: "Cross-platform" is the only one that reads like free text. Severity: LOW /

arguably not-a-violation. Listed for completeness. Remediation (optional): if these ever need to vary per product, move to per-entry fields in `portfolio.json`; otherwise leave as structured-data constants.

Not violations (false-positives explicitly cleared)

- `home.go:401,404–407` `t0r(lang,"skip","Skip to content"),`
`t0r(lang,"nav.work","Work"),`
`t0r(lang,"nav.products","Products"),`
`t0r(lang,"nav.contact","Contact")` — the English literals are `t0r` *fallbacks* only. Verified all keys (`skip`, `nav.work`, `nav.products`, `nav.contact`, `nav.portfolio`, `nav.home`) now exist for all 15 languages in `ui-i18n.json`, so `T()` returns a real translation and the fallback never fires. Data-driven; the inline comment in `i18n.go:50–58` is stale but the behavior is correct. (Recommend refreshing that comment; not a content bug.)
 - `portfolio.go:85,126,140`, `product.go:99,161` `// %s eyebrow prefix` — the `//` is a decorative monospace design motif; the text is `T(lang, ...)`. OK.
 - All `esc(T(lang,...)) / data-i18n="key" / i18nAttr()` emissions across `home.go`, `portfolio.go`, `product.go`, `shells.go` — data-driven. OK.
 - `product.go:161` eyebrow emits `esc(e.Tier)` (e.g. `helix-primary`) as visible text — it is a raw data ID shown to users (a UX nit), but it is sourced from `portfolio.json`, so it is NOT hardcoded-in-code. Out of scope here.
 - All `const` `productStyle/portfolioStyle/vasicHeadExtras/`
`mvHeadStyle/*Symbols/ *ToggleScript` — CSS, SVG icon paths, and JS. Code, not content. OK.
 - `seo.go` `robots.txt` / `sitemap` XML scaffolding, `assetLinks`, `motionScript`, `DTCG/--od-*` token names, class names, `Jekyll` `relative_url` fragments, `data-i18n/data-od-count/data-dl` attribute names, `ogLocales/ langDisplayNames/ogLocale/htmlDir` maps (locale codes + endonyms, not prose). OK.
 - `*_test.go` — expected-output assertions, not generator emissions. Out of scope (though `F6`'s dead `consts` and `F1`'s headings are indirectly asserted by `markdown_test.go` — worth checking those tests when fixing).
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Prioritized fix list

1. **F1 (HIGH)** — move `creativeHeadings` to `i18n` (`prod.head.*` in `ui-i18n.json`) so `product` section headings are translatable + reviewer-gated.

2. **F2 + F3 (MED)** — add `alt.portrait / alt.brandLogo i18n` keys; thread `lang` into `mvPortrait/vdHeroLogo` and use `T(lang,...)`.
3. **F4 (MED)** — replace literal 3 with `len(p.Tiers)`.
4. **F5 (MED)** — replace the `product`. English meta-desc fallback with an `i18n` template or the localized title.
5. **F6 (LOW)** — delete the dead `portfolioLede/portfolioSummary` consts.
6. **F7 (LOW)** — derive the footer copyright year instead of literal 2026; unify the three footer strings.
7. **F8 (LOW/optional)** — leave JSON-LD enums, or move to per-entry data if they must vary.